1) **Simple Interest Calculation**

Formula:

$$SI = (P \times R \times N)/100$$

-- Wap to calculate the simple interest based on the given principal amount, rate of interest and number of years.

Set serverouput on

```
DECLARE
   p NUMBER := &principal;
   r NUMBER := &rate;
   n NUMBER := &years;
   si NUMBER;
BEGIN
   si := (p * r * n) / 100;
   DBMS_OUTPUT.PUT_LINE('Simple Interest = ' || si);
END;
/
```

---

2) Square and Cube of a Number

--Wap to calculate the square and cube of a given number

Set serveroutput on

```
DECLARE
   num NUMBER := &number;
   sq NUMBER;
   cube NUMBER;
BEGIN
   sq := num * num;
   cube := num * num * num;

   DBMS_OUTPUT.PUT_LINE('Square = ' || sq);
   DBMS_OUTPUT.PUT_LINE('Cube = ' || cube);
END;
/
```

---

3) Product Discount Calculation

--Wap to calculate the discount amount based on quantity and price

Set serveroutput on

```
DECLARE
   pname VARCHAR2(50) := '&product_name';
```

```
   qty NUMBER := &quantity;
   price NUMBER := &price;
   discount_percent NUMBER := &discount_percent;
   total NUMBER;
   discount_amt NUMBER;
BEGIN
   total := qty * price;
   discount_amt := (total * discount_percent) / 100;

   DBMS_OUTPUT.PUT_LINE('Product Name: ' || pname);
   DBMS_OUTPUT.PUT_LINE('Total Amount: ' || total);
   DBMS_OUTPUT.PUT_LINE('Discount Amount: ' || discount_amt);
END;
/
```

---

4) Convert Feet into cm, inch and meters

1 foot = 30.48 cm
1 foot = 12 inches
1 foot = 0.3048 meters

--Wap to convert measurement in feet into centimetre's, inches and meters

Set serveroutput on

```
DECLARE
   feet NUMBER := &feet;
   cm NUMBER;
   inch NUMBER;
   meter NUMBER;
BEGIN
   cm := feet * 30.48;
   inch := feet * 12;
   meter := feet * 0.3048;

   DBMS_OUTPUT.PUT_LINE('In Centimeters = ' || cm);
   DBMS_OUTPUT.PUT_LINE('In Inches = ' || inch);
   DBMS_OUTPUT.PUT_LINE('In Meters = ' || meter);
END;
/
```

---

5) Celsius to Fahrenheit

Formula:

$$F = (C \times 9/5) + 32$$

--Wap to convert temperature from Celsius to Fahrenheit

Set serveroutput on

```
DECLARE
   c NUMBER := &celsius;
   f NUMBER;
BEGIN
   f := (c * 9/5) + 32;
   DBMS_OUTPUT.PUT_LINE('Temperature in Fahrenheit = ' || f);
END;
/
```

---

6) Generate Numbers using LOOP, FOR LOOP and WHILE LOOP

--Wap to generate number up to user input using different looping statements

Set serveroutput on

```
DECLARE
   n NUMBER := &number;
   i NUMBER := 1;
BEGIN
   DBMS_OUTPUT.PUT_LINE('Using LOOP');
   i := 1;
   LOOP
      EXIT WHEN i > n;
      DBMS_OUTPUT.PUT_LINE(i);
      i := i + 1;
   END LOOP;

   DBMS_OUTPUT.PUT_LINE('Using WHILE LOOP');
   i := 1;
   WHILE i <= n LOOP
      DBMS_OUTPUT.PUT_LINE(i);
      i := i + 1;
   END LOOP;

   DBMS_OUTPUT.PUT_LINE('Using FOR LOOP');
   FOR i IN 1..n LOOP
      DBMS_OUTPUT.PUT_LINE(i);
   END LOOP;
END;
/
```

---

7) RESULT Table + Total, Percentage and Grade

Create Table

```
CREATE TABLE RESULT (
   Rollno NUMBER PRIMARY KEY,
   Name VARCHAR2(50),
```

```
   Sub1 NUMBER,
   Sub2 NUMBER,
   Sub3 NUMBER,
   Sub4 NUMBER,
   Sub5 NUMBER,
   Total NUMBER,
   Per NUMBER(5,2),
   Grade VARCHAR2(5)
);
```

PL/SQL Block

--Wap to calculate the total, percentage and Grade of a student using result table

Set serveroutput on

```
DECLARE
   rno NUMBER := &rollno;
   s1 NUMBER;
   s2 NUMBER;
   s3 NUMBER;
   s4 NUMBER;
   s5 NUMBER;
   total NUMBER;
   per NUMBER(5,2);
   grade VARCHAR2(5);
BEGIN
   SELECT Sub1, Sub2, Sub3, Sub4, Sub5
   INTO s1, s2, s3, s4, s5
   FROM RESULT
   WHERE Rollno = rno;

   total := s1 + s2 + s3 + s4 + s5;
   per := total / 5;

   IF per >= 75 THEN
      grade := 'A';
   ELSIF per >= 60 THEN
      grade := 'B';
   ELSIF per >= 50 THEN
      grade := 'C';
   ELSE
      grade := 'F';
   END IF;

   UPDATE RESULT
   SET Total = total,
      Per = per,
      Grade = grade
   WHERE Rollno = rno;
```

```
   DBMS_OUTPUT.PUT_LINE('Total: ' || total);
   DBMS_OUTPUT.PUT_LINE('Percentage: ' || per);
   DBMS_OUTPUT.PUT_LINE('Grade: ' || grade);
END;
/
```

---

8) Gross Salary Calculation

Formula:
Gross = Basic + DA + HRA + Medical – PF

DA = 50%
HRA = 15%
Medical = 500
PF = 10%

```
--Wap program to calculate Gross salary of employee based on EID

Set serveroutput on

DECLARE
   id NUMBER := &eid;
   basic NUMBER;
   da NUMBER;
   hra NUMBER;
   medical NUMBER := 500;
   pf NUMBER;
   gross NUMBER;
BEGIN
   SELECT BasicSal INTO basic
   FROM EMP
   WHERE EID = id;

   da := basic * 0.50;
   hra := basic * 0.15;
   pf := basic * 0.10;

   gross := basic + da + hra + medical - pf;

   DBMS_OUTPUT.PUT_LINE('Gross Salary = ' || gross);
END;
/
```

---

9) Display Male Employees in HR Dept

--Wap program to display all male employees working in HR department

Set serveroutput on

Declare

```
BEGIN
   FOR rec IN (
      SELECT * FROM EMP
      WHERE Gender = 'Male'
      AND Deptname = 'HR'
   )
   LOOP
      DBMS_OUTPUT.PUT_LINE(
         rec.EID || ' ' ||
         rec.EName || ' ' ||
         rec.BasicSal
      );
   END LOOP;
END;
/
```

---

10) Delete Employee by EID

--Wap program to delete employee record using EID

SET SERVEROUTPUT ON

```
DECLARE
   id NUMBER := &eid;
BEGIN
   DELETE FROM EMP
   WHERE EID = id;

   IF SQL%ROWCOUNT > 0 THEN
      DBMS_OUTPUT.PUT_LINE('Record Deleted Successfully');
   ELSE
      DBMS_OUTPUT.PUT_LINE('Employee Not Found');
   END IF;
END;
/
```